

Test Summary — HW05 Performance Testing

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Workflow: Returning Customer Search and Order

Scenarios thực hiện

#	Scenario	JMX File	VU	Duration	Listener
D1	Load	23127464_Load_20260813.jmx	20	Ramp 60s + Hold 360s	Summary Report
D2	Stress (rerun)	23127464_Stress_20260813.jmx	10→80	Staircase 5 bậc × 60s	Aggregate Report
D3	Spike	23127464_Spike_20260813.jmx	5→50→5	Baseline/spike/recovery ~190s	View Results Tree
D4	Endurance	23127464_Endurance_20260814.jmx	20	30 phút	Response Time Graph

Endpoint groups covered

Nhóm	Endpoints	Vai trò trong workflow
Auth-heavy	POST /api/login	Bước 1: đăng nhập, lấy JWT token
Read-heavy	GET /api/products?search=, GET /api/products/:id, GET /api/orders/my-orders	Bước 2, 3, 7: tìm kiếm, xem chi tiết, lịch sử đơn hàng
Transactional	GET /api/cart, POST /api/cart, POST /api/checkout	Bước 4, 5, 6: giỏ hàng và thanh toán

Endurance threshold

20 VU / 11,5 req/s sustained 30 phút, p95 = 40 ms, error = 0,00%, CPU max = 0,452%, RAM max = 61,61 MiB.

Issues / Performance findings

#	Loại	Mô tả	Evidence
1	Performance finding	Checkout latency tăng từ 7,9 ms → 36,3 ms sau 30 phút do order/cart accumulation	D1 vs D4 JTL

#	Loại	Mô tả	Evidence
2	Functional defect	Checkout không clear cart	Source code <code>server.js:297-308</code> + runtime probe Phase A
3	Security defect	SQL injection trong search endpoint (LIKE '%\${searchQuery}%')	Source code <code>server.js:144</code>
4	Design inconsistency	Login failed +2 attempts thay vì +1; lockout 180s thay vì 30s	Source code <code>server.js:54-57</code> vs assignment spec

2.5. Demo video

Phase	Timestamp	Link
D1 Load	22:41 - 1:25:20	YouTube
D2 Stress	1:26:00 - 1:48:36	YouTube
D3 Spike	1:48:42 - 2:09:20	YouTube
D4 Endurance	2:09:30 - 3:03:17	YouTube
Full video	0:00 - 3:03:17	https://youtu.be/slTA5ErFCQ4

Repository

- GitHub: [23127464 Performance Testing HW05](#)